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Yield curve momentum

Markus Sihvonen[1,*]

[1]Bank of Finland, Research Unit, Snellmaninaukio, Helsinki, Finland

*Corresponding author: Bank of Finland, Research Unit, Snellmaninaukio, P.O. Box 160, Helsinki, Finland.
Email: markus.sihvonen@bof.fi

Abstract

I analyze time series momentum along the Treasury term structure. Yield curve momentum is primarily due to changes in the level factor of yields. Because yield changes are partly induced by changes in the federal funds rate, yield curve momentum is related to post-FOMC (Federal Open Market Committee) announcement drift. The momentum factor is unspanned by the information in the term structure today and is hence inconsistent with standard term structure, macrofinance, and behavioral models. I argue that the results are consistent with a model with unpriced longer term dependencies.

Keywords: Bond risk premia, term structure models, time series momentum, spanning.
JEL classifications: G12, E43, E47.

1. Introduction

Past returns can predict future returns (Fama 1965). Moskowitz, Ooi, and Pedersen (2012) find evidence of medium horizon return autocorrelation among a large set of asset classes. They dub this phenomenon "time series momentum."[1]

Possibly due to the focus on a broad set of asset classes, the time series momentum literature has evolved largely separately from the vast literature on term-structure modeling and bond risk premia (e.g., Fama and Bliss 1987; Ang and Piazzesi 2003; Cochrane and Piazzesi 2005). Because of this disconnect, it is, for example, not clear whether the observed return autocorrelation of government bonds is consistent with standard term-structure models.[2] This article is an attempt to study the finer dynamics of time series momentum of government bonds, or yield curve momentum, and close the gap between the two literatures.

The term structure literature features a dichotomy between variables that are *spanned* by current yields and *unspanned* variables that contain additional information useful for predicting returns (Duffee 2011; Joslin et al. 2014). The key empirical contribution of this article is to argue that past returns are spanned neither by current yields nor previously studied possibly unspanned variables.

[1] This is a growing literature, see for example, Pitkäjärvi, Suominen, and Vaittinen (2020) and Zhang (2022).
[2] Durham (2013) analyzes the performance of a duration neutral cross-sectional momentum strategy with government bonds. He argues that some its profitability can be explained by a specific affine term structure model. However, he does not address time series momentum. Asness, Moskowitz, and Pedersen (2013) study a cross-country momentum strategy with government bonds, finding that such a strategy yields positive yet fairly small returns. Brooks and Moskowitz (2017) explain bond returns using value, momentum and carry factors. However, they do not study the sources of momentum or relate the findings to the term structure modelling literature. Osterrieder and Schotman (2017) connect bond return autocorrelations with model risk parameters but do not explicitly address momentum.

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While no-arbitrage term structure models can in principle allow for unspanned variables, theoretically motivated models imply full spanning. Therefore, my results are problematic for nearly all models attempting to *explain* yield curve momentum.

My investigation starts by establishing basic properties of yield curve momentum. First, momentum is significant only for the fairly short one month lookback horizon but not for longer horizons. I therefore focus on the positive association between the current month bond return and that in the previous month. Second, I find that the term structure of momentum coefficients is downward sloping. Slope coefficients from regressing monthly bond returns on the past month return of the same maturity bond decline with the bond's maturity.

Third, I argue that yield curve momentum occurs primarily due to *changes* in the first principal component of yields, also known as the level factor. This is even though the *level* of the level factor cannot explain momentum. The associated strong factor structure in bond returns implies that most of momentum can be captured using a single return factor defined as the average past return of different maturity bonds.

Fourth, I assess the relationship between monetary policy and yield curve momentum. Because changes in the Treasury yield curve are related to changes in the federal funds target rate, yield curve momentum is partly induced by monetary policy. That is, yield curve momentum is in part driven by a drift pattern following a recent rate change by the Fed. However, because especially long maturity yields display movements unrelated to target rate changes, yield curve momentum is not identical to post-FOMC (Federal Open Market Committee) announcement drift discussed in Brooks, Katz, and Lustig (2019).

Cieslak (2018) finds strong evidence for positive rate forecast errors in months in which the Fed eases policy but insignificant results for tightening months. This might also suggest asymmetries for the momentum results. Indeed, for short maturities, momentum appears stronger following rate cuts than hikes or positive rather than negative returns. However, such differences are mainly statistically insignificant and the patterns reverse for longer maturities. Given the sampling variation in the data, it is challenging to identify differences in momentum coefficients in subsamples.

Fifth, I analyze whether past returns are spanned by current yields. Standard term structure models imply that yields are affine in a set of factors, which also determine expected bond returns. But since the yields are a simple function of the factors, controlling for sufficiently many yields is equivalent to controlling for the factors. These models can in principle generate return autocorrelation through autocorrelation in model factors. However, they imply that past returns cannot predict future returns after controlling for yields.

I find that past bond returns predict future returns also conditional on the information in the yield curve today. In fact past returns appear largely orthogonal to current yield curve factors. Hence, the spanning condition is clearly violated in the data.

Several papers (e.g., Duffee 2011; Joslin et al. 2014) have pointed that macro variables related to real activity and inflation also appear unspanned by yields. I therefore show that my results hold when controlling for macro variables, including a large panel of such variables as in Ludvigson and Ng (2009). Hence, past returns are also unspanned by macro variables.

As mentioned, while reduced form no-arbitrage models can be parametrized to include unspanned variables, theoretically motivated models imply full spanning. This includes macrofinance models characterized through investor preferences. While some of these models imply non-linear relationship between bond returns and past yields, they still imply full spanning after controlling for these non-linear relationships.

The majority of explanations offered for momentum feature deviations from full information rational expectations. Can such behavioral theories explain my findings? Not necessarily. The reason is that the current behavioral models tend to imply the same affine form for yields though the coefficients might be different from those in rational models.

The key modeling contribution of this article is to (1) construct a term structure model consistent with my empirical findings and (2) offer a theoretical interpretation for the model. Here agents do not understand that factors possess longer term dependencies and bond prices reflect this misunderstanding. The misspecification of true factor dynamics leads to a momentum pattern similar to that observed in the data. Moreover, it leads to a violation of the standard spanning condition so that past bond returns predict future returns conditional on standard yield curve factors. Finally, this misinterpretion also explains the forecast errors documented in interest rate surveys.

I explain that my term structure models are consistent with the form of bounded rationality discussed by Molavi (2019) and Molavi, Tahbaz-Salehi, and Vedolin (2021). Here an agent can only entertain models with at most a fixed number of factors and chooses a misspecified model that gives a best representation of the data. This constraint on model complexity leads agents to ignore longer dependencies in factors.

2. Data and definitions

I start by describing the data sources and variable definitions.

2.1 Bond yields and returns

I use the dataset on zero coupon US Treasury yields constructed by Liu and Wu (2021). These yields are built using a novel non-parametric method, which implies lower pricing errors compared to previous interpolation procedures. I apply a sample of end of month data between August 1971 and December 2019. Data are available for bonds with maturities up to 10 years. In the Supplementary Appendix, I show that the key results are robust to using the alternative dataset constructed by Gürkaynak, Sack, and Wright (2007) (Table 13), Bloomberg zero coupon curves (Table 13), German zero coupon data (Table 14), US and international bond indices (Section OA. 9 and Table 15) as well as Treasury bond futures data (Tables 11 and 12).

I denote the monthly continuously compounded yield of a bond (or bill) with n months until maturity by y_t^n. The logarithmic excess monthly return of maturity n bond is then given by

$$rx_{t+1}^{n} = -(n-1)y_{t+1}^{n-1} + ny_t^n - y_t^1 \tag{1}$$

Here the excess return is calculated as the monthly bond return deducted by the one month bill rate. The return between month t and any month $t+h$, $rx_{t,t+h}^n$, is given by the sum over the one period excess returns.

To save space, in most of the tables and figures, I show results for bonds with integer annual maturities rather than all the 120 different maturities. That is, these focus on bonds with maturities $12, 24, \ldots, 120$ months, that is, $1, 2, \ldots, 10$ years. Table 1 shows key descriptive statistics for these yields and excess returns.

2.2 Macro variables and other data

I construct a large panel of 135 other macroeconomic and financial variables. This contains all the variables in the FRED-MD database of McCracken and Ng (2016). Following Moench and Siavash (2022), I further add eight variables. This includes the weekly hours of production and non-supervisory employees, the Philadelphia Fed leading indicator for the US economy, the VXO index, and the measure of realized stock market volatility of Berger, Dew-Becker, and Giglio (2020). Moreover, the data contain the Bank of America Merrill Lynch MOVE bond volatility index, a measure of financial uncertainty from Ludvigson, Ma, and Ng (2021), the excess bond premium from Gilchrist and Zakrajšek (2012), and the 3-Month Treasury bill forecast from the Consensus Economics. The measures based on academic papers are extended to cover my sample period accordingly.

Table 1. Descriptive statistics for bond yields and excess returns.

	Yields (%)									
Maturity	1	2	3	4	5	6	7	8	9	10
Mean	5.06	5.31	5.50	5.69	5.82	5.96	6.06	6.14	6.22	6.27
Volatility	3.54	3.49	3.40	3.32	3.23	3.18	3.11	3.06	3.00	2.93
Skewness	0.48	0.40	0.38	0.39	0.39	0.42	0.44	0.45	0.46	0.45
Ex. kurtosis	−0.15	−0.32	−0.36	−0.37	−0.38	−0.37	−0.32	−0.31	−0.28	−0.26
Obs#	580	580	580	580	580	580	580	580	580	580
	Excess returns (%)									
Maturity	1	2	3	4	5	6	7	8	9	10
Mean	0.07	0.10	0.15	0.17	0.20	0.22	0.22	0.25	0.24	0.26
Volatility	0.44	0.83	1.18	1.51	1.80	2.08	2.33	2.61	2.86	3.13
Skewness	1.20	0.54	0.14	−0.12	−0.02	0.09	0.11	0.13	0.11	0.10
Ex. Kurtosis	17.39	13.87	8.21	5.04	4.11	4.11	3.06	2.51	2.25	2.18
Obs#	579	579	579	579	579	579	579	579	579	579

The data are monthly but the bond yields are expressed in annual form. Maturities are in years. The sample is from August 1971 to December 2019.

I also show results when controlling only for the Chicago Fed National Activity Index, used, for example, by Joslin et al. (2014), and the trend inflation measure used in Cieslak and Povala (2015). Here I apply a smoothing parameter of 0.987 in monthly updating terms to annual core inflation.[3]

Finally, I obtain the federal funds target rate and the relevant target ranges from FRED.

3. Simple regression evidence

I first consider a univariate regression of the form

$$rx^n_{t+1} = \alpha + \beta rx^n_{t-h,t} + \epsilon_{t+1} \tag{2}$$

That is, I regress the excess return of an n maturity bond in month $t+1$ on the excess return of an n maturity bond between months $t-h$ and t. When calculating excess returns, I hold maturity constant by rolling over the bond each month. I focus on lookback horizons (h) of 1, 3, 6, and 12 months. I apply Newey and West (1987) standard errors and the lag selection procedure of Newey and West (1994), which gives five lags.[4] Figure 1 illustrates the results for the slope coefficients; the full results are given in Supplementary Appendix Table 1.

The results are statistically significant for the return over the past month. However, the results for longer horizon past returns are mainly not significant. Therefore, for the rest of this article, I focus on the one month horizon. This is in contrast to Moskowitz, Ooi, and Pedersen (2012) who focus on 1 year past returns.[5] I also ignore the volatility scaling

[3] The trend inflation τ^{CPI}_t is calculated using $\tau^{CPI}_t = (1-\nu)\sum_{i=0}^{t-1} \nu^i \pi_{t-i}$, where ν is smoothing/learning parameter. Cieslak and Povala (2015) compute ν using survey data.

[4] However, there are a couple of exceptions. For the annual lookback horizon, I apply 11 lags, which is the number of overlapping observations induced by regressing monthly returns on annual returns. Note that my main specifications have no overlapping observations. For the 2000–2019 subsample regressions, shown in the Supplementary Appendix, the procedure gives four lags. Moreover, it gives three lags for the 2020–2023 post-sample regressions.

[5] Note that here the significance of 1-year past returns is somewhat better than that for 3- and 6-month past returns. However, still here only the slope coefficients for 3- and 4-year bonds are significant at the 5 percent level.

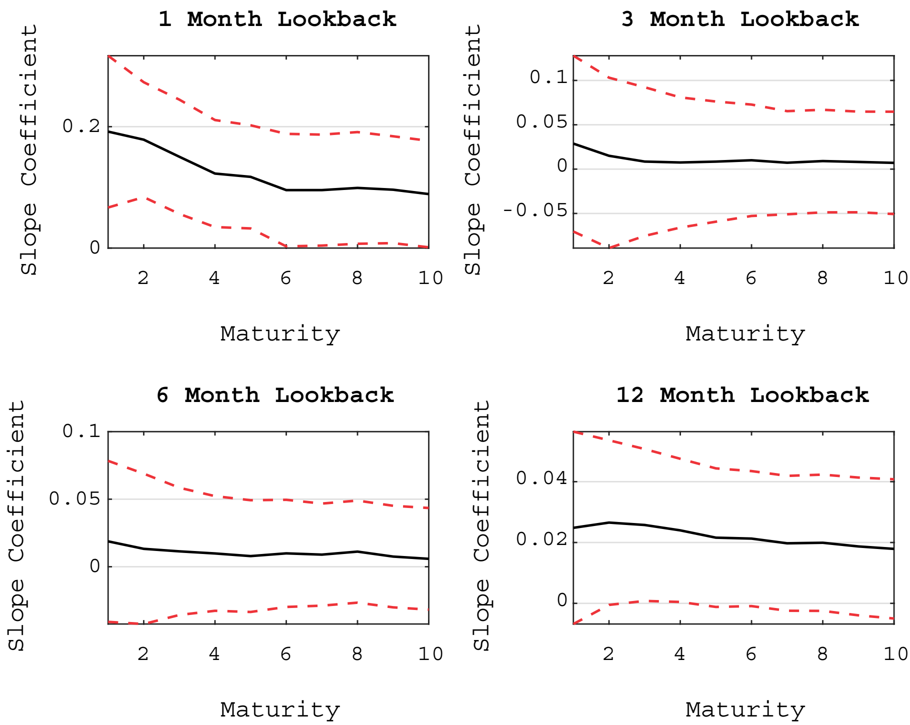

Figure 1. The figure shows the slope coefficients and the relevant 95 percent confidence bounds from regressing the monthly excess returns of different maturity bonds (years) on the past excess return for the same maturity bond for lookback horizons of 1, 3, 6, and 12 months. The sample is from August 1971 to December 2019.

applied by Moskowitz, Ooi, and Pedersen (2012) as it can induce return predictability unrelated to raw momentum in returns as discussed in Kim, Tse, and Wald (2016) and Huang et al. (2020). Moreover, in contrast to Moskowitz, Ooi, and Pedersen (2012), I avert pooled regressions due to potentially biased slope estimates and issues with calculating standard errors (Hjalmarsson 2010; Huang et al. 2020).

The regression betas decline in bond maturity. Hence, the term structure of momentum coefficients is downward sloping. This is inconsistent with one-factor interest rate models.[6]

The results for the 1-month horizon have strong economic significance as illustrated in Figure 2. It shows the mean excess returns for different maturity bonds both for the full sample and in two subsamples with positive and negative past month excess returns for the same maturity bond. The mean returns are substantially higher following positive rather than negative past month returns and the mean returns are increasing in bond maturity. The Supplementary Appendix contains additional results concerning the investment performance of simple momentum strategies.

3.1 Factor momentum

Yields and bond returns are often found to exhibit strong factor structures (e.g., Litterman and Scheinkman 1991). Hence, yield curve momentum might also be captured well using a simple factor. I next demonstrate that most of this momentum can indeed be represented by a single factor.

[6] I show this in a working paper version of the article, available on my webpage.

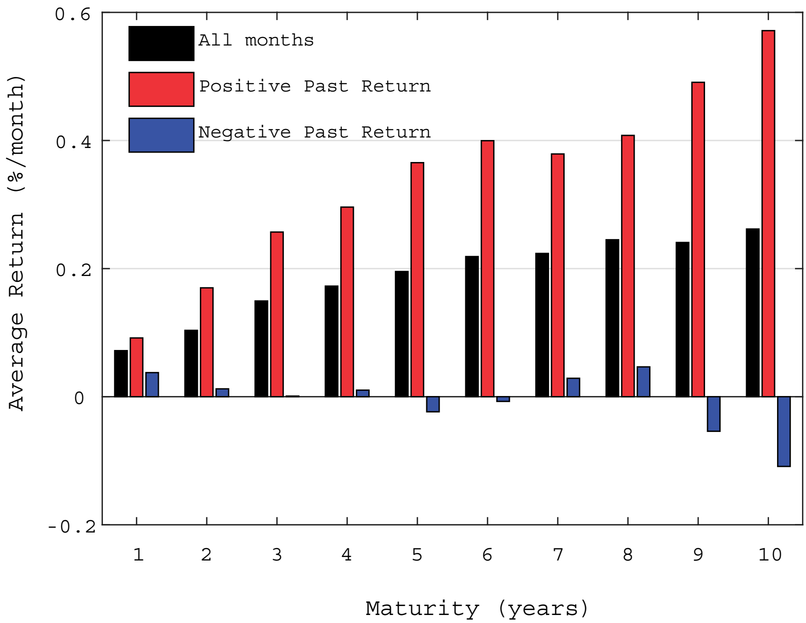

Figure 2. The figure shows the mean excess returns for different maturity bonds both for the full sample and in subsamples following positive and negative past month returns. The sample is from August 1971 to December 2019.

Let us create a simple average of the different maturity bond returns as

$$\bar{rx}_t = \frac{1}{10}\sum_{n\in N} rx_t^n, \tag{3}$$

where $N = \{12, 24, 36, 48, 60, 72, 84, 96, 108, 120\}$, that is, I apply the integer annual maturities between 1 and 10 years. I then run a regression

$$rx_{t+1}^n = \alpha + \beta\bar{rx}_t + \epsilon_{t+1} \tag{4}$$

The results are given in Table 2. Using the average of excess returns across different maturity bonds leads to only a minor loss in predictive power relative to using the past return of a bond with the corresponding maturity. For longest maturity bonds, the R^2 actually increases but this improvement is small. I confirm this overall result in the next section by showing that yield curve momentum is driven by a change in the first principal component of yields. Note that the loadings for the momentum factor are still different for returns based on different maturity bonds.

4. Sources of momentum

What is driving the results obtained in the previous section? This section derives three key results. First, yield curve momentum is due to a change in the level factor of yields. Second, these level factor changes are not spanned by current yields. Third, results are similar when controlling for macroeconomic variables.

Table 2. The results from regressing the monthly excess returns of different maturity (years) bonds on the previous month average excess return of different maturity bonds.

Mat.	α	t-value	β	t-value	R^2 (%)
1	0.06	3.58	0.04	2.37	3.16
2	0.09	2.68	0.08	2.75	2.73
3	0.13	2.84	0.09	2.57	2.04
4	0.15	2.56	0.09	2.43	1.31
5	0.17	2.43	0.11	2.45	1.22
6	0.19	2.37	0.14	2.65	1.46
7	0.19	2.14	0.15	2.59	1.42
8	0.21	2.08	0.17	2.55	1.33
9	0.21	1.85	0.17	2.45	1.17
10	0.23	1.85	0.18	2.27	1.05

The t-values are based on Newey and West (1987) standard errors with five lags. The sample is from August 1971 to December 2019.

4.1 The effect of level changes

I first revisit the question of whether yield curve momentum can be captured using a single factor. I extract the first principal component of yields using all the 120 maturities between 1 month and 10 years and consider the following regression:[7]

$$rx_{t+1}^{n} = \alpha + \beta \Delta pc_t^1 + \epsilon_{t+1} \tag{5}$$

I also decompose return autocovariance to a level change effect and a residual component. In particular, consider the contemporaneous projection:

$$rx_t^n = a + b\Delta pc_t^1 + e_t \tag{6}$$

Now we have

$$\begin{gathered}
\mathbb{C}ov(rx_{t+1}^n, rx_t^n) = \\
\mathbb{C}ov(\beta\Delta pc_t^1 + \epsilon_{t+1}, b\Delta pc_t^1 + e_t) = \\
\mathbb{C}ov(\beta\Delta pc_t^1 + \epsilon_{t+1}, b\Delta pc_t^1) + \mathbb{C}ov(\beta\Delta pc_t^1 + \epsilon_{t+1}, e_t) = \\
\underbrace{\beta\mathbb{V}ar(\Delta pc_t^1)b}_{\text{Level change effect}} + \underbrace{\mathbb{C}ov(\beta\Delta pc_t^1 + \epsilon_{t+1}, e_t)}_{\text{Residual effect}}
\end{gathered}$$

Here the third line uses the fact that ϵ_{t+1} must be orthogonal to Δpc_t^1. In a standard one-factor model, the first component would account for 100 percent of return autocovariance.

The first principal component explains roughly 98.5 percent of the variation in yields. This component is often called a level factor since it loads fairly evenly on all maturities. The average contemporaneous correlation between the change in this factor and excess bond returns is −0.95. That is, an increase in this factor is related to an upward shift in the yield curve but also to negative excess bond returns. This high correlation between bond excess returns and changes in the level factor indicates that these level factor changes will also explain a large fraction of yield curve momentum.

[7] The Supplementary Appendix further performs a carry-yield change decomposition. Excess returns depend on yield changes and a carry component. I find that autocorrelation in excess bond returns is due to autocorrelation in yield changes.

Table 3. The results from regressing the excess monthly returns of different maturity (years) bonds on the past month change in the first principal components of yields.

	Regression					Decomposition	
Mat.	α	*t*-value	β	*t*-value	R^2 (%)	Δpc_t^1 change	Other
1	0.07	4.00	−0.02	−2.47	3.41	86.32 %	13.68 %
2	0.10	3.05	−0.03	−2.86	2.59	86.37 %	13.63 %
3	0.14	3.14	−0.04	−2.57	1.85	88.24 %	11.76 %
4	0.17	2.79	−0.04	−2.29	1.07	82.38 %	17.62 %
5	0.19	2.65	−0.05	−2.22	0.94	80.78 %	19.22 %
6	0.21	2.60	−0.06	−2.39	1.17	109.91 %	−9.91 %
7	0.22	2.37	−0.07	−2.37	1.22	110.25 %	−10.25 %
8	0.24	2.30	−0.07	−2.36	1.21	103.88 %	−3.88 %
9	0.23	2.06	−0.07	−2.23	1.02	96.38 %	3.62 %
10	0.25	2.05	−0.08	−2.09	0.92	96.09 %	3.91 %

The table shows a decomposition of return autocovariance into an effect due to a change in this principal component and a residual component. The t-values are based on Newey and West (1987) standard errors with five lags. The sample is from August 1971 to December 2019.

Results from the predictability regression and decomposition are given in Table 3. The share of return autocovariance explained by level factor changes is on average 94 percent and ranges between 81 percent and 110 percent. A share of more than 100 percent implies that the residual component from a contemporaneous projection of returns on the change in the level factor is negatively associated with next month returns. The R^2 statistics in the regressions are of similar magnitude than in the plain momentum regressions, where returns are explained by past returns. Overall, I conclude that the bulk of yield curve momentum is explained by changes in the level factor.[8]

4.2 Spanning decomposition

Past bond returns can predict future bond returns either because (1) past bond returns contain information about current yield curve factors that predict future bond returns or (2) past returns contain additional information relevant for future returns. Formally, the first explanation implies that past returns are *spanned* by current yields whereas the second implies that they are not. As explained later, standard term structure models imply that the spanning condition holds so that yield curve momentum should be explained by the first channel.

To test the relative importance of the two channels, first, consider a contemporaneous regression of this month's excess bond returns on this month's principal components of yields. To save space, I make use of the finding that most of momentum can be captured using a single factor and look at the average return of different maturity bonds $\bar{rx}_t$. I apply the first five principal components of yields; including further components has minor effects on the results.

The results are given in Table 4. The regression R^2 statistic is only roughly 3 percent, while full spanning would imply a value of 100 percent. Hence, past returns are largely unspanned by yields. Most of the coefficients are also statistically insignificant, except those corresponding to the second principal component of yields.

The above results suggest that unspanned variation in returns is important to explaining yield curve momentum. I now test this result more formally by including the first five

[8] In the working paper version available on my webpage, I further consider the predictive power of changes in higher order principal components, finding weak results for most principal components. Also note that while level changes are important predictors of bond returns, the level of the level factor contains only minor predictive information for returns.

Table 4. The results from a contemporaneous regression of the average excess monthly return of different maturity bonds on the first five principal components of yields.

α	−0.11
t-value	−0.41
β_1, pc^1	−0.005
t-value	−1.64
β_2, pc^2	−0.05
t-value	−2.09
β_3, pc^3	0.12
t-value	1.34
β_4, pc^4	0.03
t-value	0.10
β_5, pc^5	0.33
t-value	0.90
$R^2(\%)$	2.82

The t-values are based on Newey and West (1987) standard errors with five lags. The sample is from August 1971 to December 2019.

principal components into the predictive regression shown in Figure 1. Figure 3 plots the slope coefficients on past returns; the full results are given in Supplementary Appendix Table 4. We can see that the past return is still significant, with corresponding slope coefficients of similar magnitude to those obtained before. These results further confirm that past returns are largely unspanned by current yields.[9]

4.3 Controlling for macro variables

Macroeconomic variables are often found to forecast bond returns on top of yields (Duffee 2011; Joslin et al. 2014; Cieslak and Povala 2015; Coroneo, Giannone, and Modugno 2016; Moench and Siavash 2022). As discussed later, this suggests these variables are unspanned by current yields. In theory, past returns might be correlated with such unspanned macro variables, which could explain why past returns themselves are unspanned. I next argue that my results are valid also when controlling for information in macroeconomic data.

To control for a large set of macro variables, I first follow an approach similar to that in Ludvigson and Ng (2009). Consider predicting bond returns using a factor model of the form:

$$m_{it} = \lambda' f_t + e_{it}$$

$$rx^n_{t+1} = \alpha' M_t + \beta' Z_t + \epsilon_{t+1},$$

where $M_t \subset f_t$. Here I posit that each macroeconomic variable m_{it} is driven by a smaller set of common factors f_t. These common macroeconomic factors then predict bond returns along with other variables Z_t.

The factors are extracted using principal component analysis.[10] The number of macro factors is determined using the IC2 criterion of Bai and Ng (2002). The criterion suggests that the macro data are well described by seven common factors, which explain about 44 percent of the variation in the data.

[9] The working paper version available on my webpage provides additional support that this holds when including more yields to the predictability regression as well as controlling for potential non-linearities.

[10] Missing observations are handled using the expectation maximization algorithm suggested by Stock and Watson (2002).

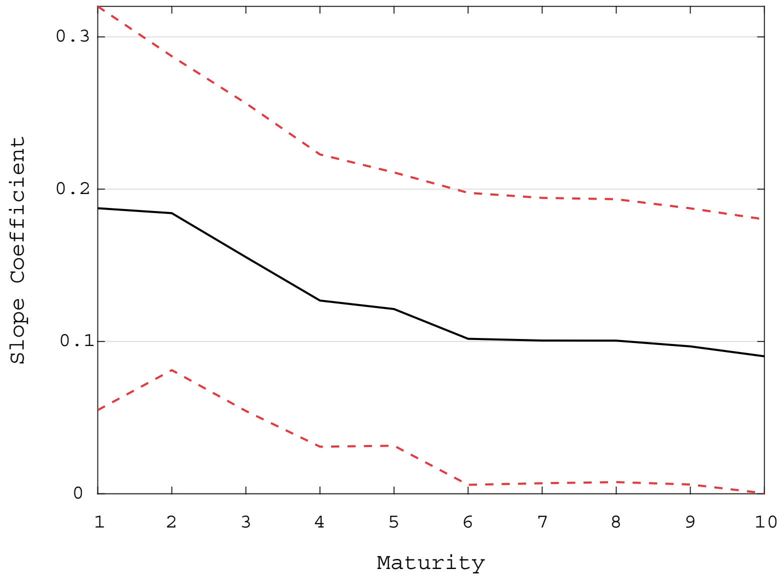

Figure 3. The figure shows the slope coefficients on the past month return for different maturity bonds when regressing monthly excess returns on their past month value and the past month's first five principal components of yields. It also plots the relevant 95% confidence bounds, based on Newey and West (1987) standard errors with five lags, for these slope coefficients. The sample is from August 1971 to December 2019.

The optimal combination of estimated factors $\hat{M}_t$ is determined using the BIC criterion. Following Ludvigson and Ng (2009), I also consider squares and cubes of the factors. Note that the optimal factors are generally different for different maturity bonds. I control for past bond returns and the first five principal components of yields, that is $Z_t = [rx_t^n, pc_t^1, pc_t^2, pc_t^3, pc_t^4, pc_t^5]$. This is in contrast to Ludvigson and Ng (2009) who only control for the Cochrane–Piazzesi factor. Note that, as in Moench and Siavash (2022), my macro panel also includes additional macroeconomic and financial variables.

The results for coefficients on past returns are given in Figure 4. The full results are shown in Supplementary Appendix Table 5. The set of selected factors includes the first, third and seventh principal components of the macro data. It also contains the squares of the first and seventh principal components as well as the cube of the fourth principal component. The chosen factors vary somewhat among the different maturities. The past returns remain significant for maturities between 1 and 5 years though less so for the longer maturity bonds. Interestingly, for short maturities, the momentum slope coefficients are also larger than before.

I also examined a version of the algorithm where the set of possible macro factors includes the lags of the first seven principal components of the macro variables. For 1 year bonds, the algorithm includes the lags of the first and second principal components but the coefficient and t-value on the past return are similar to before. The lags are not selected for the longer maturities; so, the results are exactly as before.

Principal components of macroeconomic variables lack an obvious economic interpretation. Therefore, I now also show results when instead controlling for trend inflation and the activity index. I also include lags of these macro variables, as well as the five principal components of yields.

The results for the coefficients on past returns are given in Figure 5; the full results are given in Supplementary Appendix Table 6. The slope coefficients for past returns are

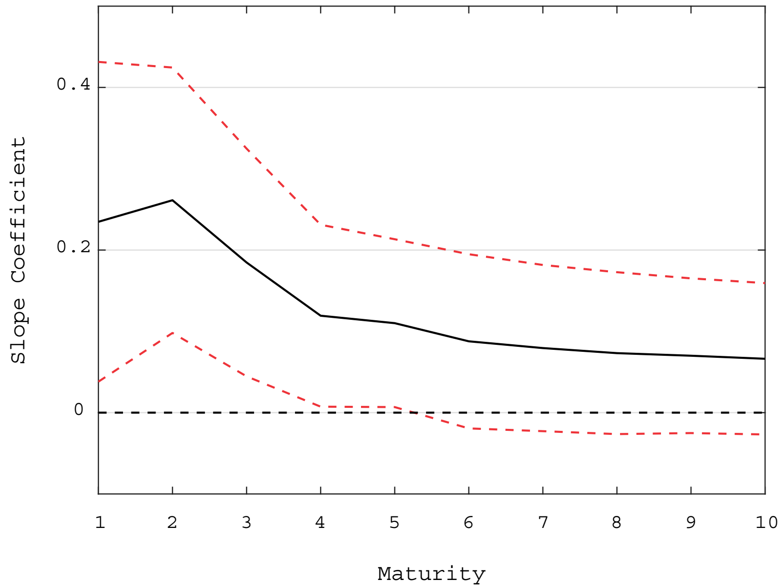

Figure 4. The figure shows the slope coefficients on the past month return for different maturity bonds when regressing monthly excess returns on their past month value, the past month's first five principal components of yields and the past month optimal set of macroeconomic predictors. The optimal macro predictors are found similarly to Ludvigson and Ng (2009). It also plots the 95% confidence bounds for these slope coefficients. These confidence bounds are based on asymptotic standard errors (Bai and Ng 2006) with the Newey and West (1987) correction with five lags. The sample is from August 1971 to December 2019.

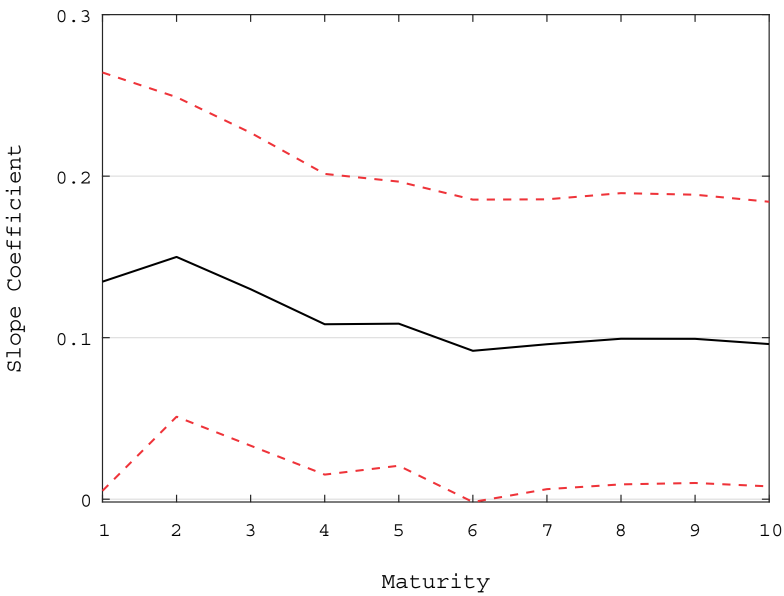

Figure 5. The figure shows the slope coefficients on the past month return for different maturity bonds when regressing monthly excess returns on their past month value, trend inflation, the national activity index, as well as the past month values of these two macroeconomic variables. It also shows the 95% confidence bounds, based on Newey and West (1987) standard errors with five lags, for these slope coefficients. The sample is from August 1971 to December 2019.

clearly significant and of similar magnitude than before. Trend inflation and its lag are also significant. Moreover, the activity index is also significant though its lag only weakly so.

I conclude that accounting for macro variables does not alter the main results of this article though these variables possibly represent additional unspanned information useful for predicting returns. However, in some specifications, the predictive content of past returns of bonds with maturities greater than 5 years appears weaker than before, while the predictive content of past returns of short maturity bonds can be higher.

5. Momentum and post-FOMC announcement drift

Because especially the short end of the yield curve tends to be tightly controlled by the Fed, yield curve momentum might be induced by policy rate changes. This is also due to recent findings related to post-FOMC announcement drift. Brooks, Katz, and Lustig (2019) find that longer term bond yields respond sluggishly to changes in the federal funds target rate.[11]

I now study this relationship using data on the federal funds target rate.[12] Figure 6 first shows the correlation between changes in yields and changes in the federal funds target rate. It does so in two samples: the full sample starting in 1982 and a subsample of months with a non-zero change in this policy rate.

Excluding months with no rate changes, this correlation is close to 0.8 at the short end of the yield curve but only around 0.3 at the long end. The decline in correlation for longer maturity bonds is natural since the federal funds rate is an overnight rate. All of these correlations are somewhat smaller in the full sample; roughly 30 percent of months included changes in the policy rate. Overall, these results suggest important dependencies between target rate and bond yield changes. However, they also indicate that not all of the yield changes can be attributed to target rate changes.

I then analyze the contribution of target rate changes to yield curve momentum using a decomposition. I project bond returns on contemporaneous changes in the target rate as follows:

$$rx_t^n = a + b\Delta FFTR_t + e_t \tag{7}$$

Using this projection, I can then decompose bond return autocovariance into an effect caused by changes in the target rate and a residual component:

$$\mathbb{C}ov(rx_{t+1}^n, rx_t^n) = \underbrace{\beta \mathbb{V}ar(\Delta FFTR_t)b}_{\text{FFR effect}} + \underbrace{\mathbb{C}ov(rx_{t+1}^n, e_t)}_{\text{Other}}, \tag{8}$$

where β is the slope coefficient from a regression of bond returns on the past month change in the target rate. The results are given in Table 5. This simple decomposition suggests that target rate changes are an important contributor to momentum for shorter maturities but less so for longer maturities. However, even for shorter maturities the contribution of target rate changes is less than 50%.

Overall, yield curve momentum is therefore connected with, but not identical to, post-FOMC announcement drift. Past month yield hikes predict low returns in the following month. These yield changes can be partly but not fully explained with same month movements in the policy rate.

11 There is a similar drift pattern in equity markets after rate changes, see Neuhierl and Weber (2018).

12 In a working paper version available on my webpage, I also consider unexpected shocks in the rate. Here the data begin later in February 1990.

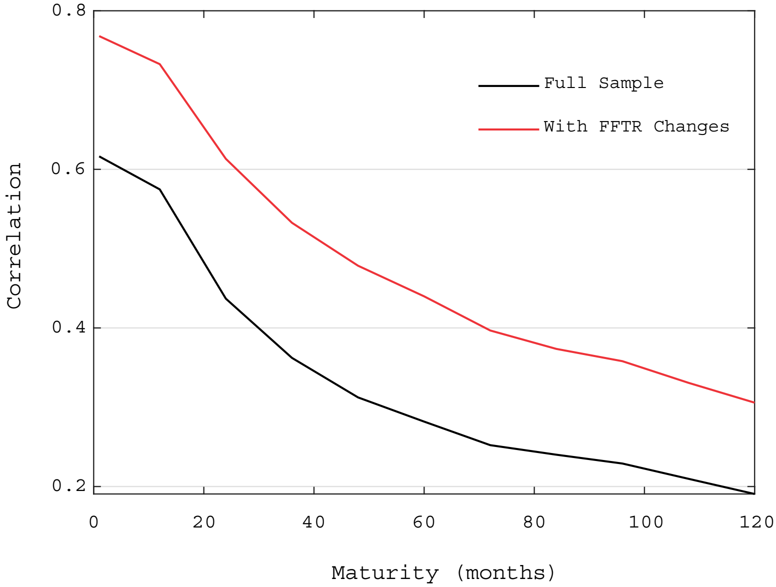

Figure 6. The figure shows the correlation between the monthly change in the federal funds target rate (FFTR) and the change in the yield of different maturity (in months) bonds in the full sample as well as in the subsample of months with non-zero FFTR changes. The sample is from October 1982 to December 2019.

Table 5. The decomposition of monthly autocovariance in excess returns of different maturity (years) bonds into a part explained by change in the federal funds target rate and a residual component.

Maturity	FFTR effect	Other
1	47.3%	52.7%
2	31.0%	69.0%
3	20.3%	79.8%
4	21.3%	78.8%
5	17.1%	82.9%
6	13.6%	86.4%
7	5.3%	94.7%
8	7.4%	92.6%
9	4.8%	95.2%
10	4.8%	95.2%

The sample is from October 1982 to December 2019.

I then more formally test the hypothesis that past returns, on top of target rate changes, contain additional information that is useful for predicting future returns. In particular, I regress the maturity-specific excess return on the corresponding past month excess return as well as the past month change in the target rate. The results are given in Table 6. The coefficients associated with the target rate change are insignificant. On the other hand, the coefficients for past returns remain significant for bonds with maturities between 1 and 5 years. While these coefficients are also insignificant for longer maturity bonds, note that here the sample is shorter than in our main specifications.

These results further support the idea that, while the two are related, yield curve momentum is not well captured by the change in the target rate. Again the logic is that past month

Table 6. The results from regressing the monthly excess returns of different maturity bonds (years) on the past month excess return for the same maturity bond as well as the past month change in the federal funds target rate.

Mat	1	2	3	4	5	6	7	8	9	10
α	0.07	0.11	0.17	0.21	0.25	0.29	0.32	0.36	0.38	0.41
t-value	5.82	4.34	4.09	3.82	3.58	3.48	3.30	3.24	3.03	3.02
β_1, rx	0.16	0.16	0.13	0.10	0.10	0.07	0.06	0.04	0.04	0.03
t-value	2.45	3.03	2.81	2.23	2.18	1.56	1.32	0.95	0.94	0.70
$\beta_2, \Delta FFTR$	−0.11	−0.17	−0.20	−0.26	−0.25	−0.18	0.01	−0.05	−0.01	−0.03
t-value	−1.56	−1.21	−0.94	−0.90	−0.71	−0.44	0.02	−0.08	−0.01	−0.04
R^2 (%)	5.75	3.88	2.57	1.58	1.27	0.65	0.36	0.20	0.19	0.11

The t-values are based on Newey and West (1987) standard errors with five lags. The sample is from October 1982 to December 2019.

returns can be only partly attributed to target rate changes. The Supplementary Appendix contains additional results concerning the post-FOMC announcement drift.

Cieslak (2018) finds evidence for asymmetries in forecast errors for Fed easings and tightenings. She argues that forecast errors for the FFR are large and negative when Fed cuts the rate but insignificant for rate hikes. This finding is further corroborated by Schmeling, Schrimpf, and Steffensen (2022). I next study whether there are also some asymmetries in the momentum results for easings and tightenings.

In particular, I consider the following specification

$$rx^n_{t+1} = \alpha + \beta_1 I(\Delta FFTR < 0) rx^n_t + \beta_2 I(\Delta FFTR > 0) rx^n_t + \beta_3 I(\Delta FFTR = 0) rx^n_t + \epsilon_{t+1} \quad (9)$$

That is, I interact past returns with three dummies, $I(\Delta FFTR < 0)$, $I(\Delta FFTR > 0)$, and $I(\Delta FFTR = 0)$, and study whether the effect of past returns is different. Rates remain unchanged in about 69% of the months, the share rate hikes and cuts is roughly equal.[13] The sample starts from 1982 when the FFTR becomes available.

The results are given in Table 7. For short maturities, the momentum slope coefficients are larger for rate cuts than for rate hikes or no rate changes. However, a Wald test cannot reject the null that the slope for rate cut months is equal to that in rate hike or no rate change months. For longer maturities, the pattern is reversed in that the slope coefficients are lower for rate cut months. For the 9-year maturity, a Wald test can actually reject the null that the slopes for rate cuts and no rate hike months are equal at the 5 percent level.

Overall, there is then some weak evidence for asymmetries for rate cut, hike, and no change months. However, overall, the results are inconclusive. In the Supplementary Appendix, I further study asymmetries in months with positive and negative returns. While slope coefficients tend to be higher in months with positive returns, the differences are mostly statistically insignificant and there are again some reversals. I also show that auto-correlated bond returns over FOMC announcement days do not explain my results.

6. Momentum and affine term structure models

How to account for the above empirical findings in a term structure model? I start by introducing a baseline term structure model before moving to the quantitative exercise.[14]

[13] Still, rates declined overall during the period as the rate cuts were larger than the corresponding hikes.
[14] The working paper version available on my webpage further discusses minimal theoretical requirements implied by the data and explains why standard models fail to explain especially the violation of the spanning condition.

Table 7. The results from regressing the monthly excess returns of different maturity bonds (years) on the past month excess return for the same maturity bond interacted with dummies for cases when the monthly change in the federal funds target rate is negative, positive, and zero.

Mat	1	2	3	4	5	6	7	8	9	10
α	0.06	0.10	0.16	0.20	0.24	0.29	0.33	0.37	0.40	0.43
t-value	4.99	3.88	3.86	3.62	3.47	3.50	3.41	3.37	3.24	3.23
$\beta_1, \Delta FFTR < 0$	0.37	0.29	0.22	0.18	0.11	−0.00	−0.08	−0.14	−0.21	−0.23
t-value	3.75	2.89	2.21	1.75	1.13	−0.01	−0.80	−1.17	−1.52	−1.47
$\beta_2, \Delta FFTR > 0$	0.08	0.12	0.10	0.06	0.04	0.01	0.00	−0.02	−0.00	−0.01
t-value	0.60	1.06	0.92	0.57	0.33	0.12	0.01	−0.18	−0.03	−0.06
$\beta_3, \Delta FFTR = 0$	0.12	0.14	0.14	0.11	0.13	0.12	0.13	0.13	0.14	0.13
t-value	1.44	2.26	2.26	1.87	2.24	2.11	2.11	2.20	2.37	2.13
$\mathbb{P}(\beta_1 = \beta_2)$	0.12	0.31	0.49	0.45	0.66	0.92	0.61	0.52	0.28	0.26
$\mathbb{P}(\beta_1 = \beta_3)$	0.82	0.88	0.80	0.72	0.47	0.39	0.33	0.27	0.27	0.29
$\mathbb{P}(\beta_2 = \beta_3)$	0.06	0.22	0.51	0.54	0.83	0.26	0.09	0.06	0.04	0.05
R^2 (%)	6.13	3.34	1.83	0.86	0.58	0.27	0.50	0.84	1.57	1.59

The table shows the *P*-values from Wald tests for the equality of coefficients. The t-values and Wald tests are based on Newey and West (1987) standard errors with five lags. The sample is from October 1982 to December 2019.

For generality, and similarly to Piazzesi, Salomao, and Schneider (2015), consider three probability measures. $\mathbb{P}$ represents objective probabilities as viewed by a rational econometrician. For simplicity, I omit this $\mathbb{P}$ symbol from expectations taken under rational beliefs. $\mathbb{S}$ expresses subjective beliefs of a representative agent. Finally, $\mathbb{Q}$ is a pricing measure defined below.

Assume that the state perceived as important for determining bond prices is an $m \times 1$ dimensional factor X_t^s. This may generally be different from the $m^e \times 1$ true state vector in the economy X_t, which can in particular include additional factors $X_t^s \subset X_t$, $m \leq m^e$. Under the subjective measure, the factor X_t^s follows:

$$X_t^s = \mu^s + \phi^s X_{t-1}^s + v_t, \tag{10}$$

where v_t is multivariate Gaussian $v_t \sim N(0, V)$. I assume that under the objective measure, the true state of the economy also follows a Gaussian VAR model with coefficients μ and ϕ.

The log nominal discount factor, expressed under the subjective measure, is a linear function of the subjective factors

$$\begin{aligned} \mathcal{M}_{t+1} &= \exp\left(-\delta_0 - \delta_1' X_t^s - \frac{1}{2}\lambda_t' V \lambda_t - \lambda_t' v_{t+1} \right) \\ \lambda_t &= \lambda_0 + \lambda_1 X_t^s \end{aligned} \tag{11}$$

I can then solve bond prices recursively using

$$p_t^n = \log \mathbb{E}_t^S(\mathcal{M}_{t+1} \exp(p_{t+1}^{n-1})) \tag{12}$$

with $p_t^0 = 0$. In this model, prices and yields take a standard affine form.

$$p_t^n = A_n + B_n' X_t^s \tag{13}$$

here

$$B_{n+1} = -\delta_1 + B_n'\phi^*, \quad A_{n+1} = -\delta_0 + A_n + B_n'\mu^* + \frac{1}{2}B_n'VB_n$$

with $A_0 = 0$, $B_0 = 0$. Here the risk neutral parameters are given by

$$\phi^* = \phi^s - V\lambda_1 \tag{14}$$

$$\mu^* = \mu^s - V\lambda_0 \tag{15}$$

These risk neutral parameters define the pricing measure $\mathbb{Q}$ under which the factor X_t^s follows a VAR process with modified parameters. This pricing measure is equivalent to the subjective measure $\mathbb{S}$.

6.1 Accounting for momentum in a term structure model

I now discuss how to account for momentum in a term structure model. I initially consider a reduced form no-arbitrage setting but later offer a potential theoretical interpretation for the approach in Section 6.3.

Consider a term structure model with five principal component factors similar to that in Adrian, Crump, and Moench (2013). That is, let $X_t^s = [pc_t^1, pc_t^2, pc_t^3, pc_t^4, pc_t^5]$. Since the model features a large set of parameters, I estimate it using linear regressions as proposed by Adrian, Crump, and Moench (2013).

Now consider the following twist. Assume that under $\mathbb{P}$, and as suggested by the data,[15] the first principal component of yields pc^1 depends also on its second lag. That is, the true state variable is $X_t = [pc_t^1, pc_t^2, pc_t^3, pc_t^4, pc_t^5, pc_{t-1}^1]$. The factor dynamics can be represented in a VAR(1) model in companion form and are detailed in the Supplementary Appendix.

I solve for the momentum betas and conditional momentum betas when controlling for the first five principal components of yields by simulating the model under the above $\mathbb{P}$ dynamics, that is when the first principal component depends also on its second lag. Figure 7 shows the resulting momentum betas along with those measured from the data. Overall, one can see that the model is able to replicate yield curve momentum in the data fairly accurately. The model also matches variation in yields reasonably well. Here it implies a root mean squared error of 1.8 bps in annual terms. As documented in the Supplementary Appendix, the model additionally captures the predictive ability of the levels of the yield curve factors.

Figure 7 also shows the momentum betas implied by the standard ACM model, where the true factor dynamics are simulated assuming the level factor does not depend on its second lag. This model implies only mild autocorrelation in returns. Moreover, the conditional momentum betas are exactly zero since the model satisfies full spanning. In the modified model, the second unspanned lag of the level factor instead explains both momentum, boosting autocorrelation relative to the standard ACM model, and why past returns are unspanned by current yields.

Adrian, Crump, and Moench (2013) also generalize their estimation approach to cover unspanned variables though do not apply the method to unspanned lags. As in Joslin et al. (2014), here the violation of full spanning occurs due to knife-edge restrictions on model parameters. Note that I instead estimate my model as if the standard ACM model is correct but simulate the model under different, more general, and realistic dynamics for the state variables. This is consistent with my bounded rationality interpretation of the model

15 See the Supplementary Appendix for the estimation results when controlling for all the five principal components.

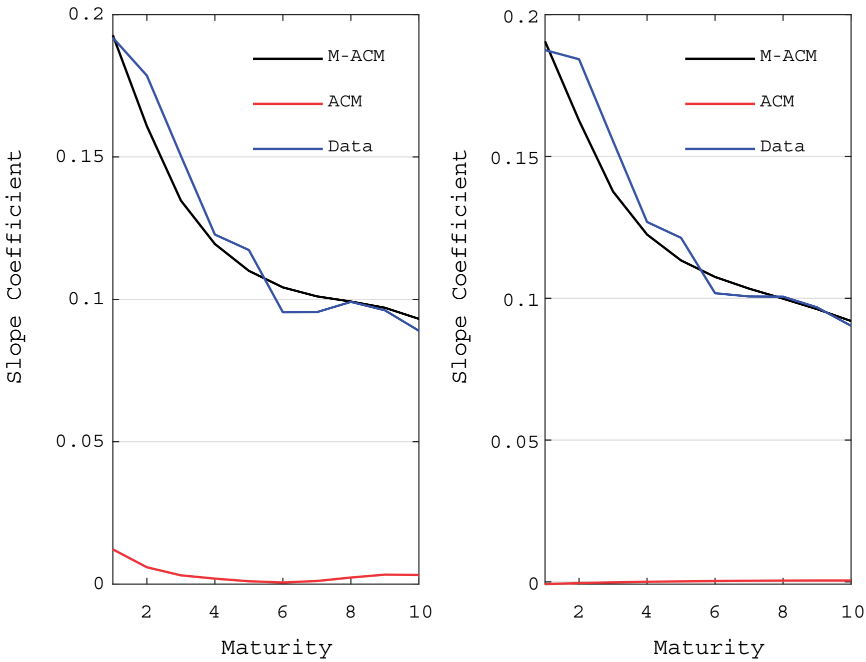

Figure 7. The figure shows the plain (left) and conditional (right) momentum coefficients observed in the data and those implied by a modified ACM model (M-ACM), where the model is simulated under a true model in which the level factor also depends on its second lag. The figure also shows the coefficients implied by the standard ACM simulated ignoring this second lag. The plain momentum slope coefficients are based on a regression of monthly excess return on their past values. The conditional slope coefficients further control for past month information in yields. Maturity is measured in years.

$\mathbb{P} \neq \mathbb{S}$, discussed later, whereas in the approach of Adrian, Crump, and Moench (2013), the agent effectively understands the true dynamics, $\mathbb{P} = \mathbb{S}$.

Alternatively, in the standard approach to unspanned variables, the change to the risk neutral measure $\mathbb{Q}$ is done from the true dynamics $\mathbb{P}$ whereas I instead make the change from the subjective measure $\mathbb{S} \neq \mathbb{P}$. These two approaches lead to a different covariance matrix of shocks. Accounting for the lag in the level factor leads to a smaller estimate of level factor residual variance.

The two approaches, however, lead to the same yield loadings B_n and therefore have identical predictions for momentum betas. The constant terms of bond prices A_n are instead generally different though in my case this difference is numerically small.

Joslin, Le, and Singleton (2013) consider an affine term structure model where the factor process has a higher order under the true than under the pricing measure. They argue that their factor dynamics are well described by a VAR(1) process under the pricing measure.[16] However, they consider a model where the factors follow a VAR(3) process under the true measure. They find that imposing no-arbitrage restrictions does not have large effects for characterizing the joint distribution of yields and macro variables.

However, they focus on the interaction between macro variables and yields and do not study momentum or any other form of return predictability. Moreover, most of their specifications include at most the first two principal components of yields. Like in Joslin et al. (2014), and unlike in this article, the change to the risk neutral measure is done from true

[16] This form, which is also assumed by Joslin et al. (2014), is convenient as it allows for rotations between latent and yield based factors.

dynamics so that the model is consistent with rational expectations models.[17] However, the approach is reduced form in that the authors do not offer an explanation for why the factor process has a higher order under the true than under the pricing measure. I instead offer a microfoundation for my model, based on the theory developed by Molavi (2019).

Finally, Feunou and Fontaine (2014) construct a model in which only the expected future values of factors are spanned by yields but their actual values are not. On the other hand, my model features a factor, whose current value is spanned but its expectation is not.

In addition to helping to understand momentum, the above results bear broad implications for term structure modeling. Institutions like central banks tend to apply reduced form term structure models with VAR(1) dynamics. These models are used, for example, to decompose bond yields to short rate expectations and term premia. My results suggest that VAR(2) dynamics might instead be more appropriate, at least when working with monthly data.

6.2 Spanning puzzle and measurement error

Cochrane and Piazzesi (2005) find that taking lags of a factor computed from forward rates can help forecast returns. They suggest that this could be explained in a model where yields are observed with error. However, in the working paper version of the paper, available on my webpage, I show that their results are largely unrelated to mine.

Duffee (2011), Joslin et al. (2014), and Cieslak and Povala (2015) find evidence that measures of inflation and real activity can help forecast bond returns on top of yields. That is, some macro variables appear to be unspanned by yields. However, Cieslak and Povala (2015) argue that the evidence is rather consistent with measurement error in yields and inflation. Similarly, Bauer and Rudebusch (2017) argue that the results of Joslin et al. (2014) are due to measurement error. Feunou and Fontaine (2014) postulate that measurement error can explain why expected inflation is not spanned by yields but not why current inflation is unspanned.

I next argue that while measurement error can possibly explain why macro variables are unspanned by yields, it does not explain why past returns are unspanned. I estimate the standard 5-factor ACM model and simulate it under the assumption that the postulated dynamics are correct. Similarly to Duffee (2011), Cieslak (2018), and Bauer and Rudebusch (2017), I introduce a normally distributed noise term to yields that is independent across maturities. I set the volatility of the error to a conservative value of 10 bps annually. This is higher than the value employed by Bauer and Rudebusch (2017) (5.8 bps) and also higher than the yield measurement error found by Liu and Wu (2021). Note that since both time $t+1$ and time t returns depend on yields at period t, both of these returns are affected by some of the same noise.

The results are given in Figure 8, which shows the simulated 5 percent (two-sided) critical values for the conditional betas. While the model implies full spanning and hence zero conditional betas, measurement error can in principle explain positive observed betas. However, the betas measured from the data are clearly above the critical values so that the momentum in the data is larger than can reasonably be accounted by measurement error. In the Supplementary Appendix, I show further results for plain momentum betas.[18]

In the Supplementary Appendix, I also find similar results for bond index and futures returns that are not subject to pricing error. In future work, robustness should be evaluated using zero coupon yields derived using the Fama and Bliss (1987) algorithm or its extensions such as that applied by Le and Singleton (2013).[19] While this method is

[17] But as argued above, this is not important from the perspective of momentum.

[18] In the working paper version available on my webpage, I also show that introducing noise to the term structure model of Cieslak and Povala (2015) does not explain my findings.

[19] The baseline Fama–Bliss data are available only for integer annual maturities and cannot be used to measure monthly returns. I therefore do not apply these data in the article.

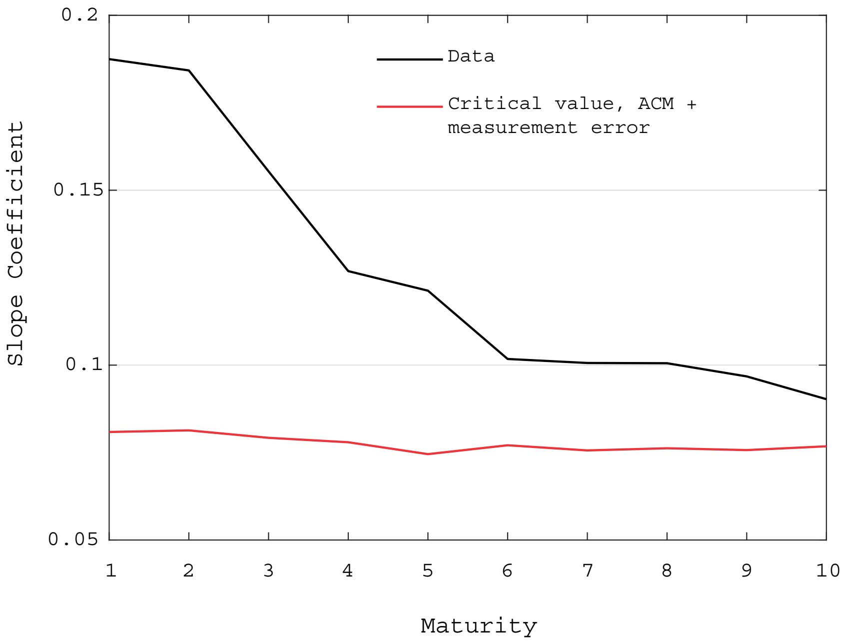

Figure 8. The figure shows the 5 percent (two-sided) critical values for the conditional momentum betas obtained by introducing measurement error to the yields obtained by simulating a standard 5-factor ACM model. It also shows conditional betas measured from the data. The conditional momentum betas represent slope coefficients on past returns from a regression of monthly excess on their past month values and the past month's first five principal components of yields. The volatility of the measurement error is 10 bps. Maturity is expressed in years.

"unsmoothed" in that the prices bonds perfectly in sample, the zero coupon yields still contain pricing error. Robustness should also be evaluated using traded cash bonds. Here trading strategy evaluation is more involved than for futures as there can be many bonds with similar tenors outstanding so discretion needs to be applied regarding which bonds to include in the trade. However, given the overall consistency of the results across different data sources, I would expect to find roughly similar results.

6.3 A bounded rationality interpretation

I have argued that the empirical results of this article are problematic for standard theories that do not naturally generate a violation of the spanning condition. But what is the economic reason that the spanning condition is not satisfied? Why are past returns important for predicting future returns but not be priced in the term structure of interest rates today? I next argue that my results are consistent with the form of bounded rationality discussed by Molavi (2019) and Molavi, Tahbaz-Salehi, and Vedolin (2021).[20]

Molavi (2019) considers a form of model misspecification in which an agent can only entertain factor models with at most d factors, where d represents the agent's sophistication. On the other hand, the agent can consider any cross-sectional relationship between model variables. Therefore, the approach captures the difficulty in dealing with time-series complexity.

[20] Molavi, Tahbaz-Salehi, and Vedolin (2021) study the asset pricing implications of this mechanism and also discuss an application to equity, but do not yield curve momentum.

More formally, the agent can only hold beliefs over the set of d-factor models:

$$F_t = \bar{\mu} + \bar{\phi} F_{t-1} + \epsilon_t$$

$$X_t^s = C' F_t$$

Here $\bar{\mu} \in \mathbb{R}^{d,1}, C \in \mathbb{R}^{d,m}$ and $\bar{\phi} \in \mathbb{R}^{d,d}$.[21] The factor might represent for example a deep macroeconomic state that is not directly observed by the econometrician. How does the agent choose the parameters $\theta = (\bar{\mu}, \bar{\phi})$? Define the Kullback–Leibler (KL) divergence of model as:

$$KL(\theta) = \mathbb{E}[-\log f^{\theta}(X_{t+1}^s | X_t^s, \ldots)] - \mathbb{E}[-\log f(X_{t+1}^s | X_t^s, \ldots)] \tag{16}$$

Here f is the true density, f^{θ} is the agent's misspecified density, and the expectations are taken under the true measure. Molavi et al. (2021) show that, among the d-factor models with a positive prior probability, the agent's posterior beliefs concentrate on the subset of models that have minimum KL divergence relative to the true data-generating process. This result is in line with the literature on learning under model misspecification (Berk 1966).

Similarly to Molavi et al. (2021), I first focus on the case where the agent can only entertain single factor models, $d = 1$. I next argue that this model gives an economic justification to a simplified version of our term structure model, detailed in the Supplementary Appendix. I then explain how the argument can be generalized to justify the multifactor term structure model used in the main part.

That is, consider a model with a factor that follows an AR(2) process. Because the factor lag enters as a state variable, this is effectively a 2-factor model. However, due to the complexity constraint, the agent can only entertain single factor models. Assume the agent can observe the factor without error and has monitored a long history.[22] Minimization of the KL-divergence amounts to a population maximum likelihood estimate of the factor dynamics, which further coincides with a population OLS estimate. Effectively, our agent learns the factor dynamics that best represent the true dynamics under a misspecified model.

Assume the representative agent's risk preference is represented by stochastic discount factor of the form 11. The bond prices are then solved using the standard price recursions. Here the state variable is one dimensional since the agent puts zero probability on models of higher dimension.[23]

This framework and the assumption that the true factor dynamics are AR(2) give an exact justification for this one-factor unspanned term structure model. Effectively, the agent cannot consider the second lag of the factor as a state variable and views the process as an AR(1). Estimation of the model can now proceed as before.

The argument could be generalized to higher dimensions, for example, to justify the multifactor unspanned term structure model in Section 6.1. Assume the agent is constrained to entertain models with at most d factors. On the other hand, assume an econometrician observes that yield levels must be described by at least d factors. Since the representative agent correctly understands all cross-sectional relationships between variables, these factors must also be taken into account in pricing. Now as the agent has exhausted its available sophistication, it cannot consider lagged values of any of these factors as a state variable, that is $m = d$ and these lags must be ignored in pricing.

21 Molavi (2019) defines the model without a constant and using a scaling matrix on the error term but this formulation is equivalent.

22 Without loss of generality $c = I$.

23 However, in a working paper version available on my webpage, I explain that the key results do not rest on the form of the stochastic discount and could also be derived for example in the case when the agent possesses mean-variance preferences.

This explanation, when applied to yield dynamics, has some similarities with the narrative discussed by Cieslak (2018). She argues that full spanning might be violated because some variables available to the econometrician are missing from the agents' information set. Here the narrative should be modified so that the agents' information set is missing higher lags of variables important for determining bond yields. This distinction is important for the results.

6.3.1 Short rate forecast errors

The model is estimated to match yield data and will therefore not match survey data exactly. These expectations can also vary depending on the survey and there could be discrepancies between the expectations of forecasters and those of market participants. However, I next use data from Consensus Economics to demonstrate that the model also does quite well in matching survey-based expectational errors.

I measure survey-based expectational errors using the consensus forecast for the 3 month Treasury bill rate a quarter ahead. The data begin in October 1989. Expectational errors are given by $FE_t^3 = y_{t+3}^3 - \mathbb{E}_t^{sur}[y_{t+3}^3]$, where $\mathbb{E}_t^{sur}[y_{t+3}^3]$ is the survey forecast. I construct a measure of the model implied *ex ante* expectational error $\mathbb{E}_t[y_{t+3}^3] - \mathbb{E}_t^S[y_{t+3}^3]$. Here the model implied subjective expectation, $\mathbb{E}_t^S[y_{t+3}^3]$, ignores the second lag of the level factor when forecasting rates, while the model implied rational expectation $\mathbb{E}_t[y_{t+3}^3]$ takes this into account.

Table 8 shows the results from explaining the survey-based expectational errors using the model implied *ex ante* expectational errors. Model implied *ex ante* errors are useful for predicting realized survey-based errors. The slope coefficient is 1.4, which is somewhat above the value of 1 necessary for unbiased forecasts. However, we cannot statistically reject the null that the slope coefficient is one. In our sample, there is still some bias in that the intercept term is statistically different from zero. This is because in this sample period, survey forecasters were expecting somewhat higher rates than those actually realized, while the model predicts no bias on average.[24]

In the model, short term return predictability arises largely due to expectational errors. I next use the survey-based forecast errors to provide evidence on this mechanism. First, Table 9 shows that survey-based forecast errors contemporaneously explain realized monthly returns. In particular, here realized returns rx_{t+1} are explained by realized forecast errors FE_{t+3}^3. There is a strong negative relationship between the two variables. Higher than expected short rates are associated with low excess bond returns.

Table 10 first shows that model-implied *ex ante* short rate expectational errors predict returns. I then extract the residual from the previous regression where returns are explained contemporaneously by survey-based expectational errors. The table then shows that *ex ante* forecast errors do not predict this residual component consistent with the idea that return predictability arises from expectational errors.

In the model, there is also a strong negative contemporaneous relationship between returns rx_t and *ex ante* expectational errors $\mathbb{E}_t[y_{t+3}^3] - \mathbb{E}_t^S[y_{t+3}^3]$. The model-implied correlation between the two is on average −0.8. High returns this month are associated with negative *ex ante* forecast errors, that is, agents predicting too high short term interest rates. These negative *ex ante* forecast errors predict high excess returns also the next month as interest rates tend to turn lower than expected. The survey data confirm that these *ex ante* forecast errors are also associated with actually realized survey-based expectational errors.

24 I have not found much evidence that these "average biases" would be contributing to momentum. For example, most of these occurred during the last 20 years of the sample, but momentum was quite weak during this period, as discussed in the Supplementary Appendix.

Table 8. The results from regressing realized survey-based short rate expectational errors on the corresponding *ex ante* expectational error implied by the modified ACM model.

	FE^3_{t+3}
α	−0.16
t-value	−3.31
β (model implied *ex ante* error)	1.41
t-value	3.95
R^2	3.14

The survey-based expectational errors are based on forecasts for the 3-month Treasury bill, 3 months ahead, obtained from Consensus Economics. The t-statistics are based on Newey and West (1987) standard errors with five lags. The sample is from October 1989 to December 2019.

Table 9. The results from a contemporaneous regression of monthly excess returns on quarterly realized survey-based short rate forecast errors.

Mat	1	2	3	4	5	6	7	8	9	10
α	0.03	0.04	0.06	0.07	0.08	0.11	0.13	0.15	0.16	0.2
t-value	4.80	2.27	2.04	1.70	1.51	1.53	1.48	1.51	1.47	1.63
β (FE^3_{t+3})	−0.22	−0.51	−0.76	−0.96	−1.09	−1.22	−1.30	−1.35	−1.40	−1.45
t-value	−9.76	−9.17	−8.78	−8.09	−7.57	−7.33	−6.74	−6.08	−5.73	−5.33
R^2	29.35	25.75	21.36	17.48	14.70	12.96	10.96	9.20	8.03	7.05

The survey-based expectational errors are based on forecasts for the 3-month Treasury bill, 3 months ahead, obtained from Consensus Economics. The t-statistics are based on Newey and West (1987) standard errors with five lags. The sample is from October 1989 to December 2019.

Table 10. The results from regressions in which next month excess returns are predicted by current month *ex ante* model generated short rate forecast error.

			Predicting returns							
Mat	1	2	3	4	5	6	7	8	9	10
α	0.07	0.12	0.19	0.23	0.27	0.31	0.34	0.37	0.39	0.44
t-value	5.54	4.40	4.36	4.09	3.89	3.84	3.69	3.58	3.46	3.53
β	−0.55	−1.18	−1.77	−2.20	−2.83	−3.04	−3.26	−3.40	−3.55	−3.50
t-value	−3.22	−2.88	−2.61	−2.41	−2.51	−2.33	−2.14	−2.00	−1.85	−1.68
R^2	2.88	2.13	1.81	1.46	1.54	1.27	1.09	0.92	0.80	0.65
			Predicting return residuals							
Mat	1	2	3	4	5	6	7	8	9	10
α ($\times 10^{-3}$)	0.1	0.1	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.4
t-value	−0.004	−0.003	−0.003	−0.002	−0.003	−0.002	−0.002	−0.002	−0.002	−0.002
β	−0.24	−0.45	−0.69	−0.85	−1.28	−1.32	−1.43	−1.49	−1.56	−1.45
t-value	−1.49	−1.22	−1.12	−1.01	−1.21	−1.08	−0.99	−0.93	−0.85	−0.73
R^2	0.77	0.42	0.35	0.26	0.37	0.28	0.23	0.20	0.17	0.12

The table shows the results from using the same variable to predict a residual component of this next month excess return. This residual component is obtained from a contemporaneous regression of excess monthly returns on realized survey-based forecast errors (see Table 9). The t-statistics are based on Newey and West (1987) standard errors with five lags. The sample is from October 1989 to December 2019. Maturities are in years.

6.4 On the possibility of rational expectations explanations

I have argued that my results are problematic for standard theoretical models that do not naturally generate violations of full spanning. However, I explained that the type of violation of spanning necessary to explain my results can be microfounded with the type of bounded rationality discussed by Molavi (2019) and Molavi et al. (2021). Could my results also be explained in a rational expectations framework? I next argue that this seems possible but providing a fully microfounded rational expectations explanation is challenging.

The reduced form model applied in this article could be justified in a rational expectations framework if the agents understand the true factor dynamics but price bonds ignoring the second lag of a key factor driving interest rates. That is, $\mathbb{S} = \mathbb{P}$ but under the risk neutral measure $\mathbb{Q}$ the factor dynamics do not feature longer lags.[25]

The literature on unspanned macroeconomic variables has featured some rational expectations narratives though not full theoretical models. Duffee (2011) argues that violations of full spanning occur if some macroeconomic variable has opposing effects for short rate expectations and risk premia and these two effects happen to net out exactly. However, it is unclear why such netting would occur for the lag of yield levels but not for their contemporaneous value. A rational expectations explanation for my results should also take the view that the expectational errors in surveys are not real or these expectations do not represent those of market participants.

As mentioned before, structural models can in principle be parametrized to generate a violation of the spanning condition. A question left for future research is to study the parameter restrictions necessary to generate momentum in a structural rational expectations model. For example, a standard version of the long run risk model takes a standard affine form in three state variables: expected consumption growth and inflation as well as conditional variance of consumption growth (Hasseltoft 2012). Momentum can in principle be generated by allowing some or all of these state variables to feature longer lags and choosing parameters so that the lags do not affect contemporaneous yields. However, since these models feature several parameters, many of which are not directly observable, evaluating whether these parameter joint restrictions are reasonable is challenging. Moreover, these parameter restrictions are knife-edge so that even small deviations from the required parameters would imply full spanning.[26]

7. Conclusion

Yield curve momentum cannot be explained by yield curve factors, as predicted by standard models. Moreover, it cannot be captured by unspanned macroeconomic variables and therefore represents an independent source of predictability. However, I show that the data are consistent with a term structure model in which agents ignore longer term dependencies in model factors.

This article bears important implications to three strands of literatures. First, by showing that past returns are an economically important yet unspanned source of predictability, it contributes to the literature on bond return predictability and risk premia. Second, it shows how momentum can be incorporated to a standard no-arbitrage setting, a useful addition to the term structure modeling literature.

Finally, the article contributes to the literature attempting to provide a theoretical explanation for momentum. When applied to government bonds, the standard theories tend to make predictions clearly violated in the data. However, I explain that momentum can be explained by the form of bounded rationality discussed by Molavi (2019) and Molavi et al. (2021). Future work should further evaluate whether rational expectations models can be extended to generate the results in this article.

25 Strictly speaking, and as explained before, this approach has somewhat different implications for the constant terms of bond yields but not for momentum slope coefficients or dynamics.

26 In particular, here the yield loadings on all lags of the standard state variables should be exactly zero.

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Supplementary material

Supplementary material is available at *Review of Finance* online.

Data availability

The key yield and macro data are publicly available and are also provided as supplementary material. The survey data are from Consensus Economics. Here many institutions have a subscription and access; the data can also be bought directly from the firm. The bond index and future data, applied in the Supplementary Appendix, can be accessed through a Bloomberg terminal.

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